

comp10002

Foundations of Algorithms

Semester Two, 2025

Welcome to Algorithmic FUN!!

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Lecture slides prepared by Alistair Moffat

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Lecturers: Alistair Moffat and Artem Polyvyanyy.



Subject Coordinator: All Academic Adjustment Plans and requests for special consideration, plus any other administrative requests, should be sent to Alistair Moffat, ammoffat@unimelb.edu.au.

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Tutors: The lecturers are supported by a team of other academic staff. You will attend a workshop each week led by one of the tutors we have selected.

Contact information and photographs for all staff are on the LMS page.

Critical information #0: The single most important thing you have to do to get the semester off to a good start is to **make sure you have friends in the class.**

Every time you enter a room over the next several weeks, be ready to chat to the other members of the group while you wait for the start of the class.

It's too late now for today, but when I come in to Wednesday's class I want to hear a babble of noise of people talking to each other.

Critical information #1: Everything is on the LMS.

Relevant reminders will be circulated each Sunday evening via an LMS announcement. *Read them carefully when they arrive!* They also remain permanently on the LMS if you need to go back and check something.

The LMS page "Modules → Weekly Schedule" will be updated each week to provide detailed instructions, and links to lecture slides and other documents.

There will be lots of other stuff build up in the LMS too: workshop solutions, assignment expectations, guides to installing C on your own computer, and so on.

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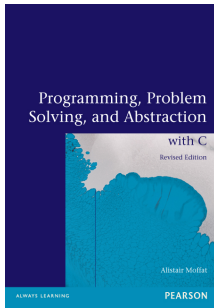
Critical information #2: Workshops will commence in **Week Two**.
There are no workshops this week.

All workshops are **on-campus**.

You may take your own computer to use if you wish.

Critical information #3: The textbook will be used extensively, including in workshops.

Programming, Problem Solving, and Abstraction with C by Alistair Moffat (second edition, Pearson, 2012); new copies are approximately \$85. An e-edition is available from the publisher's website for slightly less. There is more information at <http://people.eng.unimelb.edu.au/ammoffat/ppsaa>.



You are advised to have your own copy, new or second-hand.

Critical information #4: Attend your classes regularly and **make connections!**

Be an **active contributor**, by **interacting** with the tutors, and by **building networks** with your classmates that extend beyond the formal sessions.

Help yourself by helping others (but not by giving them your assignment solution; that won't help either of you).

Foundations of Algorithms provides further programming, now using the language C, with an emphasis on fundamental techniques and algorithms, and on software development skills.

Particular topics that will be covered include dynamic data structures, and the algorithms that manipulate them (lists, dictionaries); searching algorithms including pattern matching; and sorting algorithms.

We go “under the hood”, and build technical understanding.

You can use any C programming environment that you have access to, but might find it easiest to start in Ed Lessons, where we have set of structured exercises for you to work through.

Or, the LMS has handouts that describe how to install and use a C programming environment directly on your computer. You can then make use of command-line compilation using a terminal shell and [gcc](#) if you wish.

You'll see us using both of these options in the lectures.

Programming is “hands on” ...

The emphasis is on **you** doing programming, and learning the necessary skills in a **hands on** manner.

You need to work steadily through the semester, and write (and execute) programs throughout. You will also need to develop your knowledge of both programming techniques, and of the processes that lead to the development and analysis of algorithms.

Programming is like driving a car, you need lots of actual practice to become good at it.

There will three lectures a week.

Lectures will always be recorded. But do please **attend in person whenever you can.**

Live watching with friends will be a far better experience than lonely late-night watching by yourself.

If you don't plan to attend in person, be sure to **schedule three hours for lecture viewing in your weekly timetable**, so that you remain up to date with the content.

The workshops will start with one hour of tutorial-style interaction. The second hour will then be individual programming work, with the tutor remaining in the room through that second hour in order to answer questions.

The assigned exercises will assume that you are up to date with the lecture content. You should start working on them **before** each workshop.

Some workshops are “BYOD”. But you can use your computer in any workshop if you wish. *Remember to keep it backed up!*

Each (and every!) week:

- ▶ Three hours of lectures (live or via recordings).
- ▶ Two to three hours reading the text.
- ▶ An hour preparing for each workshop.
- ▶ Two hours attending the workshop.
- ▶ An hour of general review, perhaps in a study group.

In total, around ten hours per week per subject is required, starting immediately.

Plus the additional time spent working on assessment items.

Make a study timetable that captures all of your University commitments. **Then start following it right from Week One.**

If you have outside interests (including work) that consume more than approximately 10–12 hours per week, you are seriously jeopardizing your chances of passing.

If your outside interests cannot be restricted to less than that you should consider taking only three subjects per semester.

There are a range of mechanisms to use when you need help.

- ▶ Check the LMS for general announcements.
- ▶ Search the Ed Discussion for the same question.
- ▶ Post your query to the Ed Discussion forum. Read (and perhaps answer) other posts while you wait for a response.
- ▶ Go to the First Year Center (Level 3 of Melbourne Connect, 2:00–4:00pm daily, starting Week 2).
- ▶ Ask your tutor during (or straight after) your workshop.
- ▶ Ask straight after a lecture.
- ▶ If the issue is personal, directly email the subject coordinator, Alistair Moffat, ammoffat@unimelb.edu.au.

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comp10002 is part of the University pilot of the “PASS” program.

These are opt-in additional classes run each week by experienced tutors who have completed comp10002 recently. They aim to develop subject knowledge and skills in a fun learning environment.

The PASS classes are completely voluntary, and are in *addition* to your timetabled comp10002 workshop.

For further information see the PASS page in the LMS at “Modules → Student Support”. Anyone may attend, why not give it a try?

Your final mark is the combination of five components.

Task	When	Marks
Workshop Activities	Weeks 2–9 (best 5 of 8)	5%
Mid-Sem Test	29 Aug, 2:00pm (Week 5)	15%
Assignment 1	19 Sep, 6:00pm (Week 8)	20%
Assignment 2	17 Oct, 6:00pm (Week 11)	20%
Examination		40%

Hurdle: To pass the subject as a whole you must attain at least 22/55 (combined) in the MST and Exam, at least 16/40 (combined) in the two assignments, and at least 50/100 overall.

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All assessed work in this subject is **individual**.

We run sophisticated similarity checking software over all submissions. If you are clever enough to outsmart this software, you are also clever enough to do your own project.

The University's Academic Integrity policy may be applied if duplicate work is detected. Possible penalties include zero marks for that assessment, or even subject failure.

You will be required to submit an "Authorship Declaration" with both assignments. **Substantial mark reductions will apply if you omit it, or do not comply with it**

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Things to be done:

- ▶ Check that you can access the LMS page.
- ▶ Get hold of the textbook, [Programming, Problem Solving, and Abstraction with C](#). Start reading it.
- ▶ Confirm your workshop time and whether it is a BYOD class, and check the LMS for any late messages about changes.
- ▶ Most importantly, prepare to make some new friends; get set for a great semester; and start believing that **Algorithms are Fun!!**
- ▶ Why Fun? *Brace yourself...*